Study	All Neurometabolites	Cohen's *d*	95% CI	Weight
Al−iedani 2020		1.54	[0.62; 2.46]	0.8%
Al−iedani 2020		−2.89	[−4.04; −1.75]	0.7%
Arm 2021		0.05	[−0.58; 0.68]	1.1%
Arm 2021		−0.71	[−1.37; −0.06]	1.1%
Azevedo 2014		0.52	[0.19; 0.84]	1.5%
Azevedo 2014		0.64	[0.09; 1.18]	1.2%
Azevedo 2014		−0.06	[−0.73; 0.60]	1.1%
Azevedo 2014		0.01	[−0.32; 0.34]	1.5%
Azevedo 2014		−0.12	[−0.65; 0.42]	1.2%
Azevedo 2014		−0.39	[−1.06; 0.28]	1.1%
Cawley 2015		−0.27	[−0.87; 0.33]	1.2%
Cawley 2015		−0.26	[−0.85; 0.34]	1.2%
Cawley 2015		−0.10	[−0.69; 0.50]	1.2%
Chard 2002		−0.24	[−0.78; 0.30]	1.2%
Chard 2002		−0.96	[−1.55; −0.37]	1.2%
Confort−Gouny 1993		−0.64	[−1.43; 0.15]	1.0%
Fernando 2004		0.38	[−0.11; 0.86]	1.3%
Gao 2018		−0.59	[−1.11; −0.06]	1.2%
Huiskamp 2023		0.22	[−0.31; 0.74]	1.3%
Huiskamp 2023		−0.11	[−0.63; 0.41]	1.3%
Huiskamp 2023		0.24	[−0.36; 0.85]	1.2%
Huiskamp 2023		−0.48	[−1.10; 0.13]	1.1%
Imrich 2021		0.53	[−0.47; 1.52]	0.8%
Imrich 2021		0.25	[−0.74; 1.23]	0.8%
John 2024		−0.81	[−1.54; −0.09]	1.0%
John 2024		−0.10	[−0.79; 0.60]	1.1%
Juchem 2023		0.66	[−0.38; 1.70]	0.7%
Juchem 2023		0.22	[−0.92; 1.35]	0.7%
Kantorová 2017		0.04	[−0.49; 0.56]	1.2%
Kantorová 2022		−0.22	[−0.75; 0.31]	1.2%
Kantorová 2022		0.18	[−0.35; 0.71]	1.2%
Kara 2025		0.80	[0.22; 1.38]	1.2%
Li 2024		−0.09	[−0.52; 0.34]	1.4%
Li 2024		−0.23	[−0.66; 0.20]	1.4%
Li 2024		−0.61	[−1.04; −0.17]	1.3%
Madsen 2024		−0.31	[−0.84; 0.22]	1.2%
Madsen 2024		0.08	[−0.45; 0.61]	1.2%
Madsen 2024		−1.03	[−1.75; −0.31]	1.0%
Madsen 2024		−1.15	[−1.88; −0.42]	1.0%
Muhlert 2014		−0.42	[−1.15; 0.31]	1.0%
Muhlert 2014		−1.30	[−2.37; −0.23]	0.7%
Muhlert 2014		−0.01	[−0.73; 0.72]	1.0%
Muhlert 2014		−0.47	[−1.47; 0.54]	0.8%
Muhlert 2014		−1.47	[−2.28; −0.66]	0.9%
Muhlert 2014		−1.63	[−2.74; −0.52]	0.7%
Muhlert 2014		−1.33	[−2.12; −0.53]	1.0%
Muhlert 2014		−1.35	[−2.42; −0.28]	0.7%
Nantes 2017		0.70	[0.17; 1.22]	1.2%
Nantes 2017		0.52	[0.00; 1.05]	1.3%
Obert 2016		0.37	[−0.34; 1.09]	1.0%
Obert 2016		0.26	[−0.45; 0.97]	1.0%
Obert 2016		−0.12	[−0.88; 0.64]	1.0%
Obert 2016		0.26	[−0.50; 1.03]	1.0%
Polacek 2019		0.27	[−0.30; 0.84]	1.2%
Quadrelli 2019		0.63	[0.19; 1.06]	1.3%
Sastre−Garriga 2005		0.76	[0.23; 1.28]	1.2%
Sastre−Garriga 2005		0.18	[−0.33; 0.69]	1.3%
Solana & Martinez−Heras 2018		−0.32	[−0.76; 0.12]	1.3%
Solana & Martinez−Heras 2018		−0.14	[−0.58; 0.29]	1.3%
Srinivasan 2005		0.95	[0.22; 1.68]	1.0%
Srinivasan 2005		1.21	[0.47; 1.95]	1.0%
Srinivasan 2005		−0.23	[−1.02; 0.57]	1.0%
Swanberg 2021		−0.24	[−0.80; 0.33]	1.2%
Swanberg 2021		−0.94	[−1.57; −0.30]	1.1%
Tisell 2013		1.23	[0.50; 1.96]	1.0%
Tisell 2013		0.45	[−0.23; 1.12]	1.1%
Tisell 2013		1.13	[0.46; 1.79]	1.1%
Tisell 2013		0.41	[−0.21; 1.04]	1.1%
Vrenken 2005		0.88	[0.16; 1.60]	1.0%
Vrenken 2005		0.00	[−0.51; 0.51]	1.3%
Vrenken 2005		0.23	[−0.40; 0.87]	1.1%
Wiebenga 2015		0.07	[−0.62; 0.76]	1.1%
Wiebenga 2015		−0.11	[−0.80; 0.58]	1.1%
Wiebenga 2015		−0.12	[−0.81; 0.57]	1.1%
Wiebenga 2015		−0.35	[−1.09; 0.38]	1.0%
Wiebenga 2015		0.01	[−0.72; 0.74]	1.0%
Wiebenga 2015		0.05	[−0.68; 0.78]	1.0%
Wiebenga 2015		−0.82	[−1.54; −0.11]	1.0%
Wiebenga 2015		−0.73	[−1.44; −0.02]	1.0%
Wiebenga 2015		−1.26	[−2.01; −0.52]	1.0%
Wiebenga 2015		−0.17	[−0.90; 0.56]	1.0%
Wiebenga 2015		−0.43	[−1.17; 0.30]	1.0%
Wiebenga 2015		−0.75	[−1.51; 0.00]	1.0%
Wiebenga 2015		−0.37	[−1.07; 0.32]	1.1%
Wiebenga 2015		0.35	[−0.34; 1.04]	1.1%
Wiebenga 2015		−0.52	[−1.26; 0.22]	1.0%
Wiebenga 2015		0.54	[−0.20; 1.28]	1.0%
Xie 2025		−0.61	[−1.03; −0.20]	1.4%
Zhang 2024		−0.03	[−0.53; 0.47]	1.3%
Zhang 2024		−0.51	[−1.01; −0.00]	1.3%
Zhang 2024		−0.66	[−1.17; −0.15]	1.3%
Random effects model		**−0.10**	**[−0.22; 0.02]**	**100.0%**

Heterogeneity: I^2 = 70.8%,
τ^2 = 0.2308, p < 0.0001

−3 −2 −1 0 1 2